CSC-343: Midterm Exam

(Advanced Search (Optimization) Review Questions)

- **Show all work** for full credit. Answers without supporting steps or reasoning will receive little or no credit.
- Present your work in a **clear, logical, and organized manner**. Do not scatter steps with arrows or disconnected notes.
- **Clearly indicate your final answers** (e.g., box or highlight them).
- For proofs or problems, state all assumptions explicitly and justify each step.
- Only non-graphing calculators are permitted. No other electronic devices (including cell phones) are allowed.
- Use a pencil or erasable pen for your responses.
- For questions requiring responses that are not computational, you must use complete sentences.
- Any extraneous commentary about exam content will result in a reduced professionalism grade. If you have concerns, ask clarifying questions instead.
- Scratch paper is for your use only. It will be collected but not reviewed. Transfer all final work to this exam.
- Your responses should reflect the notation and techniques presented in class. While alternative conventions exist, consistency with course material is required for full credit.

By signing below, I certify that the work on this exam is my own without the use of external resources.

_____ _____

Printed Name Signature

Short Answer

1. When should you use optimization techniques (like hill climbing) instead of path-finding search (like A*)?

2. What is the "local optima problem" in hill climbing? Why does it occur?

3. Explain the exploration versus exploitation tradeoff in simulated annealing. How does temperature control this tradeoff?

4. What is the purpose of the cooling schedule in simulated annealing? What happens if the temperature decreases too quickly?

5. In genetic algorithms, what is the role of crossover? How does it differ from mutation?

6. Explain why elitism is commonly used in genetic algorithms. What problem does it solve?

7. How does the definition of "neighbor" affect hill climbing performance? What is the tradeoff between small and large neighborhoods?

8. Why might a genetic algorithm outperform hill climbing on a problem with many local optima?

True or **False**

1. A* search is more memory-efficient than hill climbing and genetic algorithms.

2. Hill climbing is guaranteed to find the global optimum if the search space has only one local optimum.

3. In simulated annealing, decreasing the temperature too quickly can cause the algorithm to behave like hill climbing.

4. Genetic algorithms maintain a population of solutions, which helps them explore multiple regions of the search space simultaneously.

5. In the 2-opt neighborhood for TSP, reversing a segment of the tour always produces a valid tour.

6. A larger neighborhood in hill climbing always leads to better solutions because more options are considered.

7. In genetic algorithms, a higher mutation rate always leads to faster convergence to the optimal solution.

8. Random restart hill climbing can find the global optimum with probability approaching 1 if run with enough restarts.

Other Questions

1. Consider the following optimization landscape where we want to *maximize* $f(x)$. Each state has neighbors immediately to its left and right.

State	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
$f(x)$	2	4	6	8	5	2	4	7	10	6	3	5	8	11	6	6	6	13	9	6

(a) Identify all local maxima in this landscape. Which is the global maximum?
(b) Starting from state C, trace hill climbing. Show each step and the final state reached.
(c) Starting from state G, trace hill climbing. What state does it terminate at?
(d) Starting from state P, trace hill climbing. What state does it terminate at?

2. In simulated annealing, the probability of accepting a worse move is given by:

$$P = e^{-\frac{|f(current) - f(neighbor)|}{T}}$$

where we are maximizing f, and T is the temperature.

(a) Calculate the acceptance probability when the current state has $f = 50$, neighbor has $f = 45$, and $T = 10$.
(b) Calculate the acceptance probability for the same scenario, but $T = 2$.
(c) As $T \to 0$, what does simulated annealing behave like? As $T \to \infty$?

3. For each problem below, recommend the most appropriate algorithm (Hill Climbing, Simulated Annealing, or Genetic Algorithm) and justify your choice.

(a) A phone camera engineer is adjusting two continuous parameters (lens offset and gain). A tiny change in either parameter produces a tiny, predictable change in sharpness. Evaluations are instant and you already have a decent starting setting.
(b) You must schedule 200 exams into time slots and rooms with constraints: no student conflicts, room capacities, and professor availability. There are astronomically many combinations, and you want a good feasible timetable, not necessarily the perfect one.
(c) You are optimizing a complex objective where almost every local improvement leads to a dead end (think "rugged mountains"), and a good solution requires taking several worse steps before reaching a much better region.
(d) You are calibrating a sensor by adjusting a single value, and the accuracy curve is basically a single smooth hill (one peak). Starting near the peak, you just need to climb to the top.